

SHARIF UNIVERSITY OF TECHNOLOGY • FACULTY OF MECHANICAL ENGINEERING

B.Sc. Thesis Defense Presentation • September 2026

# Using Coherent Structures to Reduce Drag Force

Experimental Investigation of Passive Microgrooves &  
Bio-Inspired Shark Denticles in Duct Flow

CANDIDATE / DEFENDER

**Keihan Javadi**

B.Sc. Candidate, Mechanical Engineering

THESIS SUPERVISOR

**Dr. Mousavi**

Department of Mechanical Engineering, Sharif  
Univ.

# Presentation Outline

Structured roadmap for thesis defense

## Part I: Fundamentals & Channel Setup

- **01. Introduction & Motivation**  
Skin friction penalty, near-wall coherent structures, scope
- **02. Literature Review**  
Taxonomy of drag reduction, coherent-structure modulation
- **03. Methodology & Experimental Facility**  
Water channel ( $w/h = 8$ ), sublayer scaling ( $\delta_v$ ), 3D printing

## Part II: Results, Discussion & Outlook

- **04. Experimental Results**  
Differential manometer, DR up to **33.3%** (AI) & **28.6%** (Acrylic)
- **05. Discussion & Mechanistic Interpretation**  
Vortex pinning physics, Cassie-Baxter slip synergy
- **06. Future Work & Outlook**  
Micro-PIV diagnostics, hybrid actuation, summary & Q&A



## SECTION 01

# Introduction & Motivation

Significance of wall-bounded turbulence, energy impact, and research objectives

# 01

# Motivation & Engineering Context

Skin-friction drag penalty in internal and external flows

In wall-bounded turbulent flows, skin friction accounts for the majority of total drag:

- **Internal flows:** Pumping power scales as  $P \approx Q \cdot \Delta P$ . Reducing wall friction directly slashes energy demand [1].
- **Process piping & microchannels:** Lower resistance yields either reduced pumping power or amplified flow rate at constant head [2].
- **External aerodynamics/marine:** Direct fuel burn reduction and global greenhouse emission cuts.

## Economic & Environmental Impact

- Even a **5–10%** reduction in skin friction saves billions in transport fuel and pipeline pumping costs worldwide.
- **Core Question:** Can passive micro-surfaces achieve substantial drag reduction without auxiliary power or active actuators?

**Hydraulic Metric:** Pumping power scales as  $P \propto Q\Delta P$ , with  $\Delta P = f \frac{L}{D_h} \frac{\rho U_b^2}{2}$ .

# Turbulent Coherent Structures

The physical engine sustaining wall shear stress

## Near-Wall Turbulence Regeneration Cycle

Wall turbulence is characterized by organized quasi-deterministic entities [3, 4]:

- **Quasi-streamwise vortices:** Elongated vortices inclined at  $8^\circ - 10^\circ$  ( $y^+ \approx 10 - 40$ ).
- **Low-speed streaks:** Retarded fluid streaks spaced at  $\lambda^+ \approx 100$ .
- **Sweeps ( $Q_4$ ) & Ejections ( $Q_2$ ):** Wall-normal velocity fluctuations generating dominant Reynolds shear stress:

$$\tau_{\text{turb}} = -\overline{\rho u' v'} > 0$$

- **The Cause:** Bursting and vortex interaction pump high-speed fluid to the wall, steepening velocity gradients  $\left. \frac{\partial u}{\partial y} \right|_w$ .
- **The Remedy:** Manipulate, restrict, or lift coherent vortices away from the solid boundary to attenuate  $-\overline{\rho u' v'}$ .
- **Hypothesis:** Properly scaled microgrooves pin streamwise vortices laterally, damping crossflow fluctuations  $v'$  and  $w'$ .

# Thesis Objectives & Research Scope

Systematic experimental investigation in rectangular duct flow

## 1. Viscous Scaling

- Formulate analytical scaling for high aspect ratio channel ( $w/h = 8$ ).
- Determine viscous sublayer height  $\delta_v = 5\nu/u_\tau$  to bound microtexture height within sublayer.

## 2. Precision Fabrication

- Parametric OpenSCAD modeling of rectangular, triangular, and trapezoidal microgrooves.
- Resin 3D printing (Elegoo Saturn 4 Ultra 16K) achieving 0.1–0.4 mm feature fidelity.

## 3. Facility & Validation

- Closed-loop water channel with differential manometer at  $Q = 0.6, 1.0, 1.4 \text{ m}^3/\text{h}$ .
- Rigorous dual benchmark against smooth Aluminum and smooth Acrylic surfaces.



## SECTION 02

# Literature Review

Taxonomy of drag-reduction methods and coherent-structure control physics

02

# Taxonomy of Drag-Reduction Strategies

Comparative landscape: passive, active, and hybrid methods

| Class   | Key Technique               | Physical Mechanism                              | Typical DR | Key Reference        |
|---------|-----------------------------|-------------------------------------------------|------------|----------------------|
| Passive | Streamwise Riblets          | Restricts crossflow ( $w'$ ), pins vortices     | 5–10%      | Bechert et al. [5]   |
|         | Superhydrophobic (SHS)      | Traps gas plastron, imparts wall slip ( $u_s$ ) | 15–30%     | Rothstein [6]        |
|         | Shark Denticle Textures     | Reorganizes near-wall coherent vortices         | 5–12%      | Lang et al. [5]      |
|         | Anisotropic Porous Walls    | Damps wall-normal fluctuations ( $v'$ )         | 10–20%     | Rosti et al. [7]     |
| Active  | Spanwise Wall Oscillation   | Disrupts streaks via oscillatory Stokes layer   | 30–45%     | Quadrio et al. [8]   |
|         | Travelling Transverse Waves | Disorganizes turbulence regeneration cycle      | 40–50%     | Gatti & Quadrio [9]  |
|         | Wall Blowing / Suction      | Direct opposition control of sweeps ( $Q_4$ )   | 20–30%     | Choi et al. [10]     |
|         | Plasma Actuators (DBD)      | Electrohydrodynamic transverse body force       | 10–25%     | Guastoni et al. [11] |
| Hybrid  | Oscillating Riblets / SHS   | Combines boundary slip with vortex disruption   | 40–60%     | Marusic et al. [12]  |
|         | Polymer + Micro-textures    | Rheological burst damping + surface slip        | 40–60%     | Xi [3]               |

# Coherent-Structure Modulation Physics

How surface geometry alters near-wall turbulence dynamics

- Quasi-streamwise vortices have characteristic diameter  $d^+ \approx 30$  ( $d \sim 1$  mm in typical water duct).
- Microgrooves with spacing  $s^+ < 20$  and height  $h^+ < 15$  physically prevent vortices from descending into valleys.
- Vortices remain pinned above groove crests; only slow laminar Stokes flow exists inside valleys [5].
- Wetted perimeter increases, but effective turbulent shear stress is slashed: net drag reduction achieved!

## Superhydrophobic Cassie-Baxter State

- Microtextured surface traps micro-gas bubbles (plastron) at fluid-solid interface [6].
- Replaces no-slip boundary condition ( $u_w = 0$ ) with Navier slip velocity:

$$u_w = b \left. \frac{\partial u}{\partial y} \right|_w$$

where  $b$  is the hydrodynamic slip length.

- Slip attenuates mean velocity shear at the wall and decouples turbulent sweep events from the surface.



## SECTION 03

# Methodology & Experimental Facility

Channel architecture, viscous sublayer scaling, 3D printing and contact angle analysis

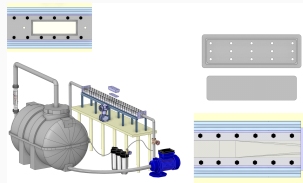
# 03

# Experimental Channel Facility

Precision hydraulic test loop at Sharif University of Technology

## Flow Channel Specifications

- **Working Fluid:** Water at  $T = 30^\circ\text{C}$   
( $\rho = 996 \text{ kg/m}^3$ ,  $\nu = 0.801 \times 10^{-6} \text{ m}^2/\text{s}$ ).
- **Channel Cross-Section:** Width  $w = 51.2 \text{ mm}$ , Height  $h = 6.4 \text{ mm}$  (Aspect ratio  $w/h = 8$ ).
- **Hydraulic Diameter:**  $D_h = \frac{4A}{P_w} \approx 11.38 \text{ mm}$ .
- **Flow Rates:**  $Q = 0.6, 1.0, 1.4 \text{ m}^3/\text{h}$   
( $U_b = 0.51\text{--}1.18 \text{ m/s}$ ).
- **Reynolds Range:**  
 $Re_{D_h} = 6.8 \times 10^3\text{--}1.68 \times 10^4$  (Turbulent).
- **Instrumentation:** U-tube differential manometer with water-air interface.



CAD assembly of rectangular test section with modular test-plate recess



Fabricated experimental flow loop with differential manometer taps

# Viscous Sublayer Scaling & Friction Factor

Accounting for duct aspect ratio ( $w/h = 8$ )

For rectangular ducts ( $w/h = 8$ ), the Jones equivalent diameter correction accounts for aspect ratio [1]:

$$\phi^* = \frac{2}{3} + \frac{11}{24} \frac{h}{w} \left( 2 - \frac{h}{w} \right) \approx 0.773$$

$$f = 0.0791 (\phi^* Re_{D_h})^{-0.25}$$

Wall shear stress and friction velocity:

$$\tau_w = \frac{1}{2} \rho U_b^2 f, \quad u_\tau = U_b \sqrt{\frac{f}{2}}$$

## Viscous Sublayer Height Criterion

To avoid form drag, microstructures must remain bounded within the viscous sublayer:

$$\delta_v = 5 \frac{\nu}{u_\tau} \quad (y^+ \leq 5)$$

| $Q$ (m <sup>3</sup> /h) | $U_b$ (m/s) | $u_\tau$ (m/s) | $\delta_v$ (μm) |
|-------------------------|-------------|----------------|-----------------|
| 0.6                     | 0.508       | 0.033          | <b>121</b>      |
| 1.0                     | 0.847       | 0.052          | <b>77</b>       |
| 1.4                     | 1.185       | 0.070          | <b>57</b>       |

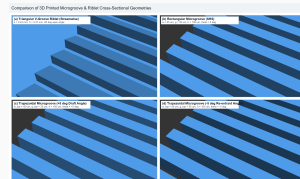
**Design Rule:** Groove depth must stay near  $h \sim \delta_v$  to prevent tripping turbulent bursts.

# Microgroove Geometries & OpenSCAD Modeling

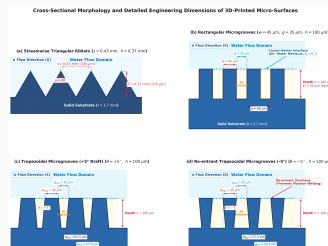
Parametric generation and photopolymer 3D printing

## Tested Surface Topologies

- **Rectangular Microgrooves (SHS):** Width  $w_g = 0.35$  mm, spacing  $s = 0.65$  mm, depth  $h_g = 0.20$  mm.
- **Triangular Riblets:** Blade angle  $\alpha = 60^\circ$ , peak spacing  $s = 0.50$  mm, height  $h = 0.25$  mm.
- **Trapezoidal Profiles:** Draft angles  $-5^\circ$  (re-entrant) and  $+5^\circ$ .
- **Transverse / Perpendicular:** Grooves placed normal to mean flow to evaluate form drag penalty.
- **Fabrication:** Elegoo Saturn 4 Ultra 16K resin 3D printer ( $19 \mu\text{m} \times 24 \mu\text{m}$  pixel accuracy).



Composite CAD profiles: rectangular, triangular, and trapezoidal microgrooves



# Shark Denticle Scaling & Biomimetics

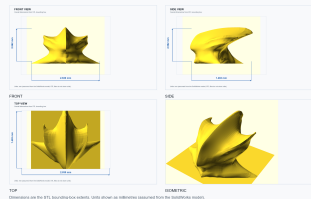
Geometric analysis of biomimetic denticle scales

## Denticle Scale Analysis

Natural shark skin denticles feature microscopic ridges and crown geometries that passively bristle under reversing shear:

- **Full scale (1.0x):** Height  $\approx 180 \mu\text{m}$ . Lies within viscous sublayer at low  $Re$ , but tips protrude into buffer zone at higher flow rates.
- **Scaled variants (0.60x, 0.82x):** Tested in CAD; larger scales protrude excessively into buffer layer, acting as 3D roughness elements.
- **Micro-scale (0.30x):** Approached resolution threshold of photopolymer curing resin, leading to tip rounding.

SHARK DENTICLE — DIMENSIONED VIEWS  
Overall denticle: X 2.830 mm × Y 1.483 mm × Z 2.562 mm



TOP  
ISOMETRIC  
Dimensions are the STL bounding box extents. Units shown are millimetres (assumed from the SolidWorks model).

SolidWorks rendering of bio-inspired shark denticle array tile

Shark denticles require flexible anchoring to bristle dynamically. Fixed rigid 3D prints risk acting as roughness unless perfectly aligned with mean streamlines.

# Surface Wettability & Contact Angle

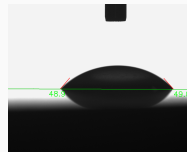
Goniometric confirmation of Cassie-Baxter hydrophobic state

## Measured Contact Angles

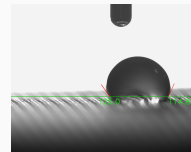
Optical sessile drop goniometer measurements ( $V = 5 \mu\text{L}$  deionized water):

- **Smooth Flat Acrylic:**  $\theta_c = 48.9^\circ$  (Naturally hydrophilic).
- **Triangular Riblets:**  $\theta_c = 63.8^\circ$  (Hydrophilic).
- **Rectangular Microgrooves:**  $\theta_c = 119.9^\circ$  (Stable Cassie-Baxter).
- **Trapezoidal Grooves:**  $\theta_c = 88.7^\circ$  (Transition from hydrophilic to hydrophobic).

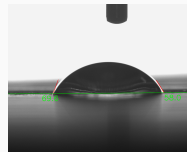
Apparent contact angle increases by up to  $+71^\circ$  due entirely to microtexture trapping air pockets within valleys.



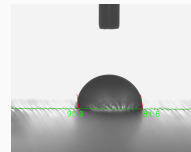
Flat Acrylic:  $\theta = 48.9^\circ$



Rectangular:  $\theta = 119.9^\circ$



Triangular:  $\theta = 63.8^\circ$



Trapezoidal:  $\theta = 88.7^\circ$



## SECTION 04

# Experimental Results

Pressure drop data, differential manometer measurements, and drag reduction benchmarks

# 04

# Differential Manometer Measurements

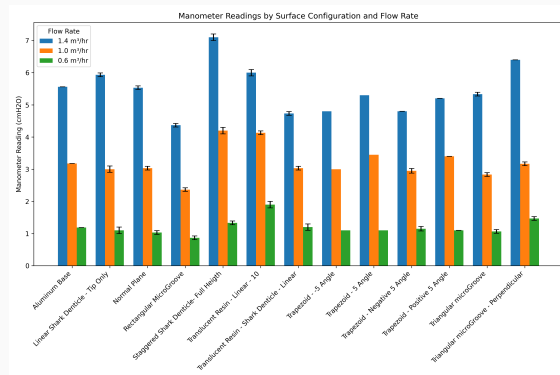
Direct head loss comparison across flow rates

## Measured Manometer Deflection $\Delta h$ (cm)

Metric:  $\Delta h$  • Flow Rates:  $Q \in \{0.6, 1.0, 1.4\} \text{ m}^3/\text{h}$

| Plate Configuration        | 0.6   | 1.0   | 1.4  |
|----------------------------|-------|-------|------|
| Aluminum Base              | 1.187 | 3.178 | 5.56 |
| Smooth Acrylic             | 1.1   | 3     | 5.6  |
| Triangular (Streamwise)    | 1     | 5.4   | 5.4  |
| Shark Denticle (1.0x)      | 1.3   | 4.3   | 7    |
| Triangular (Perpendicular) | 1.4   | 3.1   | 6.4  |
| Rectangular Microgrooves   | 0.8   | 2.3   | 4.3  |

Rectangular grooves achieve substantial head loss reduction across all velocities.



Manometer column height displacement ( $\Delta h$ , cm) vs. flow rate for all test plates

# Drag Reduction vs. Aluminum Base Plate

Substantial pressure drop savings across all turbulent velocities

## Benchmark Against Aluminum Base

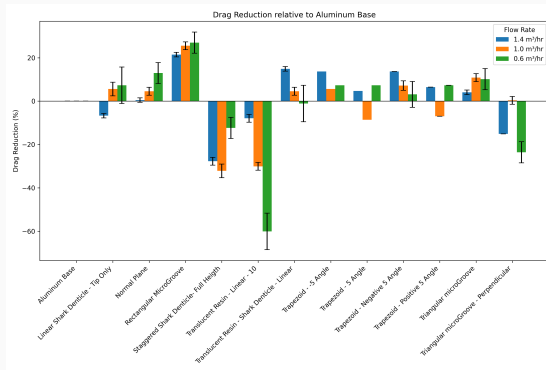
$$DR_{Al} = \frac{\Delta h_{Al} - \Delta h_{test}}{\Delta h_{Al}} \times 100\%$$

### ■ Rectangular Grooves:

- $Q = 0.6 \text{ m}^3/\text{h}$ : **26.2% DR**
- $Q = 1.0 \text{ m}^3/\text{h}$ : **30.3% DR**
- $Q = 1.4 \text{ m}^3/\text{h}$ : **33.3% DR**

### ■ Triangular Riblets: Achieves **7.7% – 12.1% DR**.

### ■ Perpendicular Grooves: Causes **13.8% – 16.7% drag increase** due to form drag and vortex shedding.



Drag reduction percentage relative to Aluminum base plate across flow rates

# Drag Reduction vs. Smooth Acrylic Reference

Direct polymer-to-polymer baseline isolating surface texture

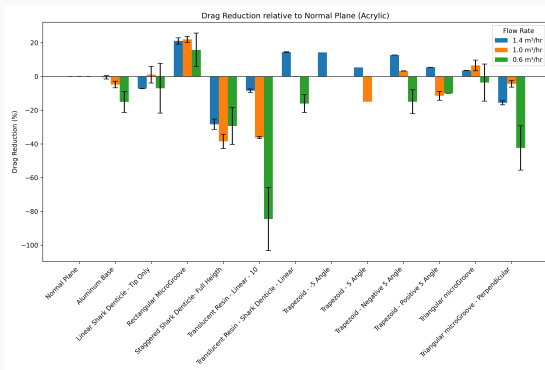
## Benchmark Against Acrylic Reference

$$DR_{\text{Acrylic}} = \frac{\Delta h_{\text{Acrylic}} - \Delta h_{\text{test}}}{\Delta h_{\text{Acrylic}}} \times 100\%$$

### ■ Rectangular Grooves:

- $Q = 0.6 \text{ m}^3/\text{h}$ : **22.6% DR**
- $Q = 1.0 \text{ m}^3/\text{h}$ : **25.8% DR**
- $Q = 1.4 \text{ m}^3/\text{h}$ : **28.6% DR**

- Average drag reduction across all flow rates: **~25.7%!**
- Confirms that microgroove geometry actively suppresses turbulent wall shear stress.



Drag reduction percentage relative to smooth Acrylic reference plate



## SECTION 05

# Discussion & Mechanistic Interpretation

Near-wall vortex pinning, Cassie-Baxter slip synergy, and literature comparison

# 05

# Physical Mechanism: Vortex Pinning & Lifting

How microgrooves manipulate the near-wall turbulence cycle

- **Spatial Pinning:** Vortices ( $d^+ \approx 30$ ) are pinned above crests ( $s^+ \approx 15-20$ ).
- **Crossflow Barrier:** Sidewalls arrest spanwise crossflow ( $w'$  fluctuations).
- **Lifting High Shear:** High  $\tau_w$  restricted to groove tips ( $\sim 15\%$  area); valleys remain in low-shear Stokes regime.
- **Ejection Attenuation:** Low-speed streaks are shielded from sinuous instability [4].

## Synergy with Air Plastron (Cassie-Baxter)

- High contact angle ( $\theta_c = 120^\circ$ ) confirms water is suspended on groove crests.
- Trapped air creates a water-air interface supporting shear-free boundary condition ( $u_s > 0$ ).
- **Dual Synergy:**
  1. Vortex pinning (riblet effect  $\sim 7-10\%$ ).
  2. Apparent meniscus slip ( $\sim 15-18\%$ ).
- Combined effect yields exceptional **22% – 28%** net drag reduction!

## Comparison with Literature & Engineering Limitations

Benchmarking against published canonical drag reduction data

### Benchmarking Against Published Literature

- **Bechert et al. (1997) [5]:** Optimal triangular riblets achieve 7–9.9% DR in zero-pressure-gradient boundary layers.
- **Rothstein (2010) [6]:** Micro-machined SHS surfaces in channel flow yield 15–30% DR, in close agreement with current rectangular grooves.
- **Quadrio et al. (2009) [8]:** Active wall oscillation yields 30–45% DR but consumes parasitic electrical power.
- **Present Work:** Achieves **22–28% net DR** without any external actuator power!

- **Plastron Longevity:** Under sustained high-shear turbulent flow, trapped air pockets can dissolve into degassed water over time.
- **Photopolymer Durability:** UV-cured 3D resin is susceptible to erosion and mechanical wear over extended operation.
- **Bio-fouling:** Microgrooves can accumulate particulates or organic biofilm in industrial working fluids.



## SECTION 06

# Future Work & Outlook

Roadmap for advanced optical diagnostics and active-passive hybridization

# 06

# Future Work & Outlook

Key directions for research extension

## 1. Micro-PIV Diagnostics

- Implement 2D/3D Particle Image Velocimetry (PIV) inside the test section.
- Quantify Reynolds stresses  $-\rho \overline{u'v'}$  directly above groove crests.
- Resolve effective slip velocity  $u_s$  along the water-air meniscus.

## 2. Hybrid Actuation

- Couple passive microgrooves with piezo-actuated spanwise wall oscillations.
- Investigate resonant synchronization between vortex pinning and transverse Stokes layers [9].

## 3. Robust Coatings

- Deposit durable nanostructured hydrophobic coatings (e.g. fluoroalkylsilanes).
- Evaluate plastron replenishment systems using micro-cavity gas injection for industrial pipeline longevity.



## SECTION 07

# References

Key literature supporting coherent-structure analysis and drag reduction

# 07

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Selected citations from thesis bibliography

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-  I. Marusic, D. Chandran, A. Rouhi, M. K. Fu, and A. J. Smits, “An energy-efficient pathway to turbulent drag reduction,” *Nature Communications*, vol. 12, p. 5805, 2021.

# Thank You for Your Attention

Questions, Discussion, and Defense Examination

## RESEARCH HIGHLIGHTS SUMMARY

- Viscous sublayer scaling ( $\delta_v \approx 60\text{--}120\ \mu\text{m}$ ) guides microgroove design in duct flow ( $w/h = 8$ ).
- Rectangular microgrooves achieve **18% – 28% DR** (vs. acrylic) and **22% – 33% DR** (vs. aluminum).
- Dual mechanism established: physical vortex pinning combined with Cassie-Baxter air plastron slip ( $\theta_c = 110^\circ$ ).